

Brady A. Pettengill

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Computational and Data Sciences student at George Mason University building practical software for data analysis, physics simulation, security, and everyday use. Focused on solving real-world problems and learning new tools.

EDUCATION

Accelerated B.S./M.S. Computational and Data Sciences

Expected May 2027

Minor Computer Science

George Mason University

Advanced Diploma, Herndon High School

June 2022

EXPERIENCE

State Farm Insurance

June 2025 – Sep. 2025

IT Intern

Vienna, VA

- Diagnosed and resolved hardware, software, and networking issues in a fast-paced office environment.
- Supported system and network setup, ensuring consistent configuration across workstations.
- Identified recurring technical issues and contributed to process improvements that reduced resolution time.

George Mason Cyber Team

Sep. 2023 – Present

CTF Competitor

Fairfax, VA

- Compete in NCL and collegiate CTFs, focusing on web, crypto, and reverse-engineering challenges.
- Apply tools such as Burp Suite, Wireshark, Nmap, Ghidra, and GDB to identify and exploit vulnerabilities.
- Collaborate with teammates to solve real-world-style cybersecurity problems within strict time limits.

Nando's PERi-PERi

Dec. 2021 – May 2024

Restaurant Manager

Reston, VA

- Oversaw financial performance, inventory management, marketing, and hiring/scheduling staff.

FIRST Robotics FTC Team

Aug 2021 – June 2022

Software Team Lead

Herndon, VA

- Organized our team to successfully design, implement, and test software for our robot.
- Developed software for both manual control and autonomous operation.

Herndon High School AFA CyberPatriot Team

Sep. 2020 – June 2022

Founder and President

Herndon, VA

- Founded HHS CyberPatriot team “CyberHornet,” reaching national semifinals in both years of competition.
- Developed our Linux sub-team, introducing members to the use and hardening of Linux-based systems.
- Worked extensively with bash scripting to automate standard competition procedures.

PROJECTS

Adaptive Cruise Control: Basic adaptive cruise control using computer vision with Python and OpenCV, communicating with the car via CAN bus and integrating with the existing cruise control through a PID controller.

Messaging App: Desktop chat platform with client-server functionality, written in Java using Swing and MySQL.

N-body Simulation: Gravitational N-body sim written in C++ with OpenMP, std::thread, and serial backends to benchmark CPU parallelization. Exposed to Python via pybind11 for real-time visualization.

Utility Website: A collection of Node.js/Express + MongoDB web apps with a Bootstrap/HTML frontend, including a URL shortener and pastebin exposed via REST APIs and the UI.

SKILLS

Languages: Python, C, C++, R, MATLAB, JavaScript, Java, SQL, Swift, Bash

Libraries: NumPy, Pandas, Scikit-learn, OpenCV, MongoDB, Node.js, Express

Tools/Platforms: Git, Linux, Docker, NGINX, Cloudflare, Wireshark, Ghidra, GDB, Nmap

Soft Skills: Team collaboration, problem-solving, adaptability, technical communication, leadership